Hints: Abstract algebra – 3. Permutations

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The cycles of σ are forced: they are the orbits of $\langle\sigma\rangle$, and on each orbit σ acts as one cycle. Disjoint cycles commute, so $\sigma^k = \mathrm{id}$ iff $c^k = \mathrm{id}$ for every cycle c, i.e. iff every cycle length divides k.

Hint 2. Conjugacy: $\tau(i_1 \dots i_k)\tau^{-1} = (\tau(i_1) \dots \tau(i_k))$. So conjugate permutations have the same cycle type. Conversely, two permutations of the same type are conjugate by the τ that sends one cycle to the other, cycle by cycle.

Hint 3. The largest order in S_n is $g(n) = \max \operatorname{lcm}(\lambda_1, \dots, \lambda_k)$ over partitions of n. You may assume the parts are prime powers with distinct primes. For $n = 10$ the maximum is $2 \cdot 3 \cdot 5 = 30$. For large n, $\ln g(n) \sim \sqrt{n \ln n}$ (Landau): use primes up to about $\sqrt{n \ln n}$.

Problem 2

Hint 1. Show that multiplying σ by a transposition (ab) changes the number of cycles by exactly ± 1: it merges the two cycles if a, b lie in different cycles and splits the cycle if they lie in the same one. It also changes the parity of inv: an adjacent swap changes inv by ± 1, and any transposition is a product of an odd number of adjacent swaps.

Hint 2. Write σ as a product of transpositions starting from id. Both $(-1)^{\mathrm{inv}}$ and $(-1)^{n-c}$ flip at every step, so they agree. The second is clearly multiplicative: a product of k transpositions has sign $(-1)^k$, so the sign is a homomorphism.

Hint 3. The reversal $k \mapsto n + 1 - k$ has all $\binom{n}{2}$ pairs as inversions, so its sign is $(-1)^{n(n-1)/2}$. This is $+1$ iff $n \equiv 0, 1 \pmod 4$.

Problem 3

Hint 1. π fixes 0, and $\mathbb{F}_p^*$ is cyclic of order $n = p - 1$ (generator g). Writing $x = g^k$, the permutation becomes $k \mapsto 3k$ on $\mathbb{Z}/n$, which is a bijection because $3 \nmid n$.

Hint 2. Use the Chinese remainder theorem: $\mathbb{Z}/n \cong \mathbb{Z}/2^a \times \mathbb{Z}/m$ with m odd, and multiplication by 3 acts on each factor. For $\sigma \times \tau$ acting on $X \times Y$, $\operatorname{sgn} = \operatorname{sgn}(\sigma)^{|Y|} \operatorname{sgn}(\tau)^{|X|}$. So the sign is that of $k \mapsto 3k$ on $\mathbb{Z}/2^a$.

Hint 3. If $a = 1$ (that is, $p \equiv 3 \pmod 4$), this map is the identity on $\mathbb{Z}/2$: even. If $a \geqslant 2$, count its cycles using $\operatorname{ord}_{2^j}(3)$ ($= 1, 1, 2$ for $j = 0, 1, 2$ and 2^{j-2} for $j \geqslant 3$): there are $3 + 2(a - 2)$ cycles, so $2^a - c$ is odd and the map is odd.

Problem 4

Hint 1. Write $\frac{1}{\nu+1} = \int_0^1 x^\nu \, dx$. Then the sum is $\int_0^1 \sum_\pi \sigma(\pi) x^{\nu(\pi)} \, dx$.

Hint 2. $\sum_\pi \sigma(\pi) x^{\nu(\pi)}$ is the determinant of the matrix with x on the diagonal and 1 elsewhere, i.e. $\det((x-1)I + J) = (x-1)^{n-1}(x - 1 + n)$ (the eigenvalues of J are $n, 0, \dots, 0$).

Hint 3. Integrate $(x-1)^n + n(x-1)^{n-1}$ from 0 to 1.

Problem 5

Hint 1. Use linearity of expectation with indicators: k is fixed with probability $1/n$. For cycles, build σ by the "closing a cycle" construction (or count): the k-th step closes a cycle with probability $1/k$, so the expectation is H_n.

Hint 2. Derangements by inclusion–exclusion: $D_n/n! = \sum_{k=0}^{n}(-1)^k/k! \to e^{-1}$.

Hint 3. Local maxima: an end position is a local maximum with probability $1/2$, an interior one with probability $1/3$ (the largest of three random values). For $n \geqslant 2$ the answer is $2 \cdot \frac{1}{2} + \frac{n-2}{3} = \frac{n+1}{3}$.